

UPH012: BIOPHYSICS AND BIOMATERIALS

L	T	P	Cr
3	0	0	3.0

Course Objective: This course introduces the concepts of chemical binding, stability and transport phenomenon to the students. Introduction to various biomaterials for specific biomedical applications is also given.

Syllabus

Chemical binding: Biomolecules, chemical bonding, interatomic potentials for strong and weak bonds, ionization potential, electron affinity, electronegativity, variation in properties with bonding character.

Thermodynamics and diffusion Kinetics: Free energy, internal energy, concept of equilibrium, stability and metastability, basic thermodynamic functions, statistical nature of entropy, kinetics of thermally activated process, Fick's law of diffusion, solution to Fick's second law and its applications, Kirkendall effect, atomic model of diffusion.

Biomaterials: Classification: Metals, Ceramics, polymers, Composites, Hydrogels, Bioresorbable and Biodegradable materials, Properties of biomaterials: Physical, Thermal, Electrical and Optical, Surface properties, Structure-property relationship of biological materials.

Novel Biomaterials: Hydrogels, Self-assembling peptides, Metallic implant materials (stainless steels, Co-based alloys, Ti based alloys), Ceramic implant materials (aluminum oxides, hydroxyapatite glass ceramics), Polymeric implants, polymers for drug delivery, In-vitro and in-vivo applications of biomaterials

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Explain various characteristics of materials based on interatomic potential plots
2. Analyse basic kinetic reactions in living systems by applying governing laws of thermodynamics.
3. Analyse the suitability of the material to be used as a biomaterial.
4. Compare the performance of different biomaterials in biomedical applications

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text Books

1. Rodney Cotterill, BIOPHYSICS An Introduction, John Wiley and Sons (2002).
2. B. D. Ratner, A. S. Hoffman, F. J. Schoen and J. E. Lemons, Biomaterials Science, Second Edition: Wiley Science (2004).
3. Temenoff J.S. and Mikos A.G., Biomaterials: The intersection of Biology and Materials Science, Pearson, (2009).
4. V. Raghavan, Introduction to Materials Science and Engineering; PHI, Delhi, 2005.

Reference Books

1. J. Breme, R.Thul and C. J. Kirkpatrick, Metallic Biomaterial Interfaces Wiley (2008).
2. L. Hench and J. Jones, Biomaterials, Artificial Organs and Tissue Engineering (Woodhead Publishing in Materials (2002).

Evaluation Scheme

Sr. No	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-45
3	Sessionals	30

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UCS421 Ethics and Risk Mitigation in AI for Engineers

Objective: Empowering engineers to navigate ethical and societal implications of AI in safety-critical domains

Modules G Content

1. Introduction to AI Ethics

Understanding the distinction between ethics and compliance is foundational for AI engineers. While compliance ensures adherence to regulations and standards, ethics delves deeper into principles guiding the responsible use of AI. Engineers will explore the core principles of fairness, accountability, transparency, and privacy, emphasizing their relevance in decision-making processes.

2. Bias in Engineering AI Models

Bias in engineering applications can manifest in various forms, such as calibration bias in sensors or the use of biased datasets during model training. Engineers will study real-world examples to understand how biases can lead to safety system failures or discrimination in access to services. This module equips participants to recognize, mitigate, and manage biases effectively.

3. Data Privacy & Governance

With the increasing integration of AI into sensitive domains like smart grids and health sensors, data privacy and governance are crucial. Participants will examine the Indian DPDP Act and GDPR, focusing on their relevance to engineering IoT systems. The goal is to understand how to handle sensitive data responsibly while maintaining compliance with global standards.

4. Explainable AI (XAI)

Explainable AI bridges the gap between black-box models and interpretable systems. Engineers will learn about tools such as SHAP and LIME, which provide insights into AI decision-making processes. Practical applications will include explaining AI-based decisions in process control systems, ensuring transparency and trust in high-stakes engineering environments.

5. AI in Critical Systems

Critical systems such as autonomous vehicles, robotics, and power grids demand rigorous risk management. Engineers will explore risk categories specific to these applications, emphasizing redundancy and fallback mechanisms. The module highlights the necessity of robust systems capable of mitigating risks and maintaining operational safety.

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6. Safety-by-Design

Safety-by-design principles entail proactive measures to ensure the reliability of AI systems. Engineers will investigate fail-safe models and strategies to combat adversarial attacks on control systems. Furthermore, the module will underscore the importance of engineering responsibility in deploying AI, ensuring that safety remains the highest priority.

7. Governance and Standards

AI governance frameworks, such as IEEE AI ethics and NIST standards, provide a structured approach to ethical deployment. Engineers will study these frameworks and learn how to create AI audit templates tailored to engineering applications. This module fosters a systematic approach to maintaining accountability and compliance.

8. AI Policy G Case Resolution

Real-world simulations will challenge participants to create AI policies for smart plant operations, addressing ethical dilemmas and practical constraints. Engineers will evaluate case studies of ethical failures to understand lessons learned and apply them to future engineering scenarios.

Practicals

- Hands-on exercises are integral to this course, enabling engineers to apply theoretical knowledge in practical settings:
- Bias detection using SHAP: Participants will analyze sensor data to identify and mitigate biases effectively.
- Privacy impact assessment: Engineers will use templates to evaluate privacy implications in smart home data.
- AI audit checklist creation: Developing comprehensive checklists for AI-driven drone navigation to ensure safety and compliance.

Case Studies

- The course incorporates comprehensive case studies to illustrate the real-world implications of ethics and risk mitigation in AI:
- Boeing 737 MAX: Analyzing software bias and identifying gaps in risk mitigation strategies.
- Tesla autopilot: Exploring failures in AI systems and the importance of robust data governance.
- AI in nuclear reactor control: Insights from IAEA on the role of ethical AI in maintaining safety in critical applications.

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Evaluation

- The assessment framework is designed to evaluate both theoretical understanding and practical application:
- Scenario-based short answers: Accounting for 30%, this segment tests engineers' ability to respond to ethical dilemmas in AI contexts.
- Ethics audit: Another 30% focuses on evaluating engineers' capacity to critically assess and audit AI systems.
- Final case report and group discussion: Constituting 40%, this segment promotes collaborative learning and comprehensive evaluation of case studies.
- By the end of this course, engineers will be equipped with the knowledge and tools necessary to implement AI systems that prioritize ethics, mitigate risks, and uphold societal trust in safety- critical applications.

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